Math 421

Wednesday, December 3

Announcements

- Drop-In Hours
  - **Today** 12-1 pm VV 311
  - Thursday 9-11 am VV B205
- MLC
  - Proof Table: M-Th 3-7:30 VV B227
- Homework 10 due Friday (last HW!)

Another Corollary of the MVT

Corollary 29.7. Let f be a differentiable function on an interval (a, b).

if $x_1 < x_2$ then $f(x_1) < f(x_2)$

1. If $f'(x) > 0$ for all $x \in (a, b)$, then f is strictly increasing.
2. If $f'(x) < 0$ for all $x \in (a, b)$, then f is strictly decreasing.

Proof of 1: Consider $x_1, x_2 \in (a,b)$ such that $a < x_1 < x_2 < b$. since $[x_1, x_2] \subseteq (a,b)$, f is continuous on $[x_1, x_2]$ and differentiable on (x_1, x_2). By MVT there exists $x_0 \in (x_1, x_2)$ such that

$$f'(x_0) = \frac{f(x_2) - f(x_1)^{\;+\;-}}{x_2 - x_1 \;+} \begin{matrix} > 0 \\ < 0 \end{matrix}$$

since $x_1 < x_2$, $x_2 - x_1 > 0$ therefore, since $f'(x_0) \overset{<0}{>} 0$, $f(x_2) - f(x_1) \overset{<0}{>} 0$ and thus $f(x_2) \underset{<}{>} f(x_1)$.

Motivating Question

Q: Is the derivative of a differentiable function always continuous?

<u>NO!</u>

☆ the only type of discontinuity a derivative can have is an essential discontinuity.

↳ not a hole, a jump or an asymptote.

Example

Let

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Then

uses product & chain rule

$$f'(x) = \begin{cases} -\cos \frac{1}{x} + 2x \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

use limit definition & squeeze theorem.

Intermediate Value Property for Derivatives

Theorem 29.8. Let f be a differentiable function on (a, b). Then $f'(x)$ has the intermediate value property on (a, b). That is:

If $a < x_1 < x_2 < b$, and if c lies between $f'(x_1)$ and $f'(x_2)$, there exists (at least one) x_0 in (x_1, x_2) such that $f'(x_0) = c$.

☆ even though we can't apply the IVT to the derivative – $f'(x)$ still has the intermediate value property.

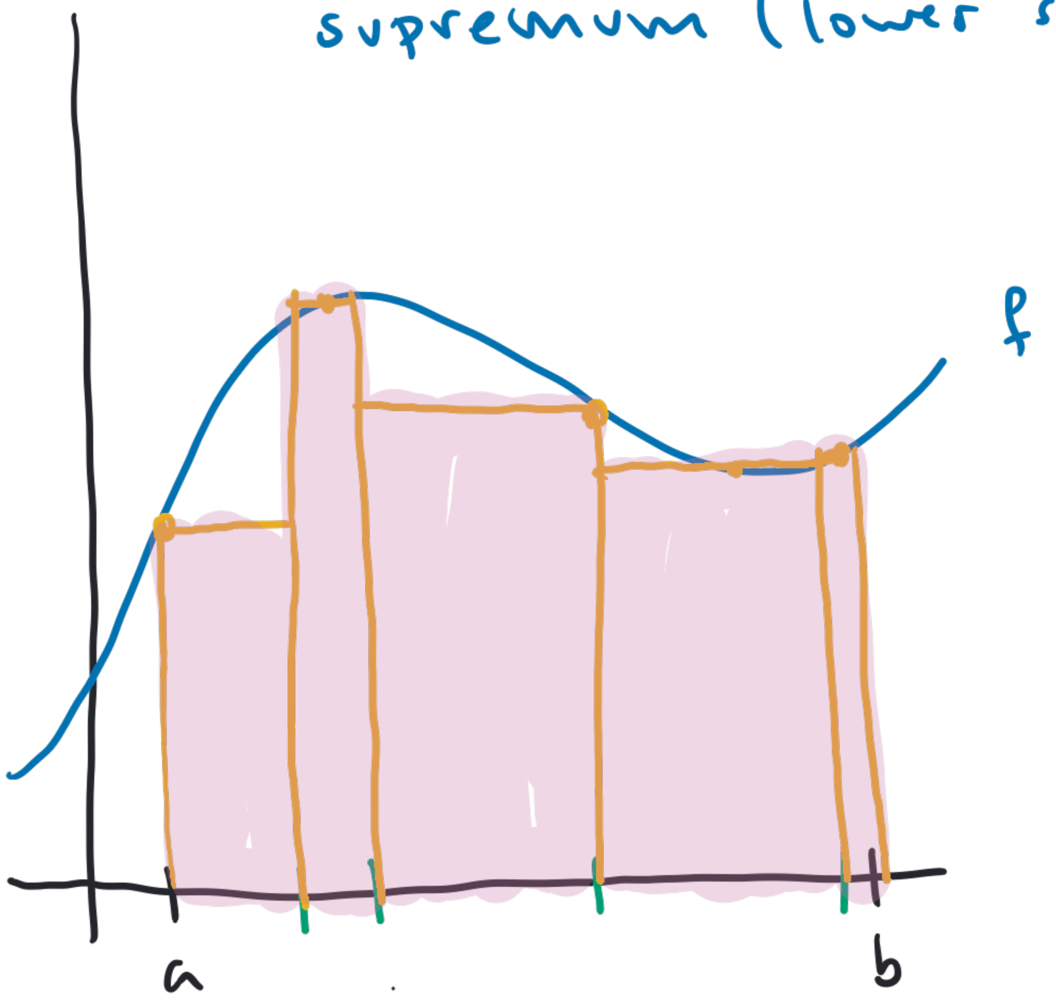
Chapter 6: Integration

Section 32: The Riemann Integral

Integration – Informal Idea

To get exact value:

infimum (upper sums) = supremum (lower sums)



1. Divide [a,b] into subintervals to form widths of rectangles. ← partitions

2. Choose heights of the rectangles. ← sup or inf

3. Calculate area of rectangles & sum ← upper sum or lower sum

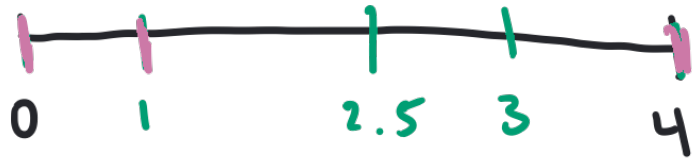
$[0,4]$

$R = \{0,4\}$

$P = \{0,1,2.5,3,4\}$

$Q = \{0,1,4\}$

$[0,1], [1,2.5], [2.5,3], [3,4]$

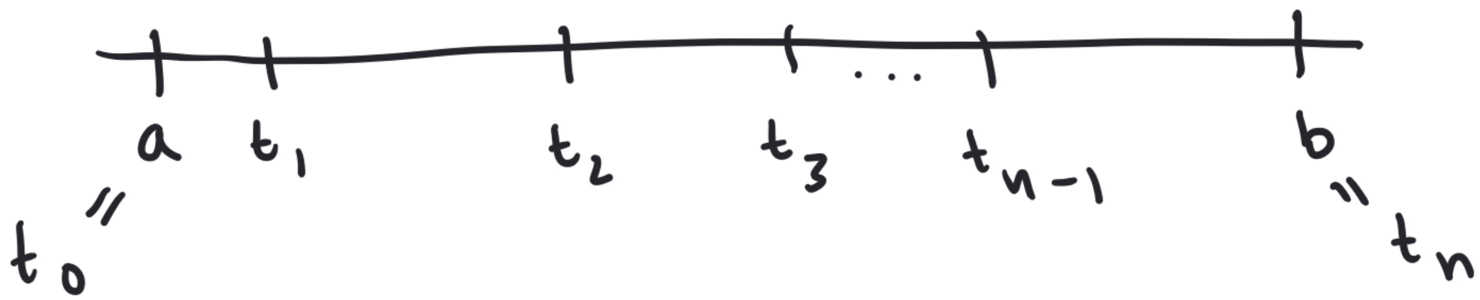


$[0,1], [1,4]$

Partitions

Def. Let $a < b$. A **partition** of the interval $[a,b]$ is a finite collection of points in $[a,b]$, $P = \{t_0, t_1, \dots, t_{n-1}, t_n\}$ such that

$$a = t_0 < t_1 < \cdots < t_{n-1} < t_n = b.$$



Upper & Lower Sums

may not be continuous.

Def. Let $f : [a, b] \to \mathbb{R}$ be bounded and $P = \{t_0, t_1, \dots, t_{n-1}, t_n\}$ be a partition of $[a, b]$. The **upper and lower (Darboux) sums** for P are defined by

$$U(f, P) = \sum_{k=1}^{n} \underbrace{M_k}_{\text{heights}} \underbrace{(t_k - t_{k-1})}_{\text{widths of subintervals from } P} \qquad M_k = \sup\{f(x) : x \in [t_{k-1}, t_k]\}$$

$$L(f, P) = \sum_{k=1}^{n} m_k (t_k - t_{k-1}) \qquad m_k = \inf\{f(x) : x \in [t_{k-1}, t_k]\}$$

Divide & Conquer

Let $f(x) = x^2$. On the interval $[-1,4]$ find the upper and lower Darboux sums for the given partitions:

1. $Q = \{-1,0,2,4\}$
   a) Upper sum
      $1(1) + 4(2) + 16(2) = 41$
   b) Lower sum
      $0(1) + (0)(2) + 4(2) = 8$
2. $P = \{-1,4\}$
   a) Upper sum
      $16 \cdot (5) = 80$
   b) Lower sum
      $0 \cdot (5) = 0$